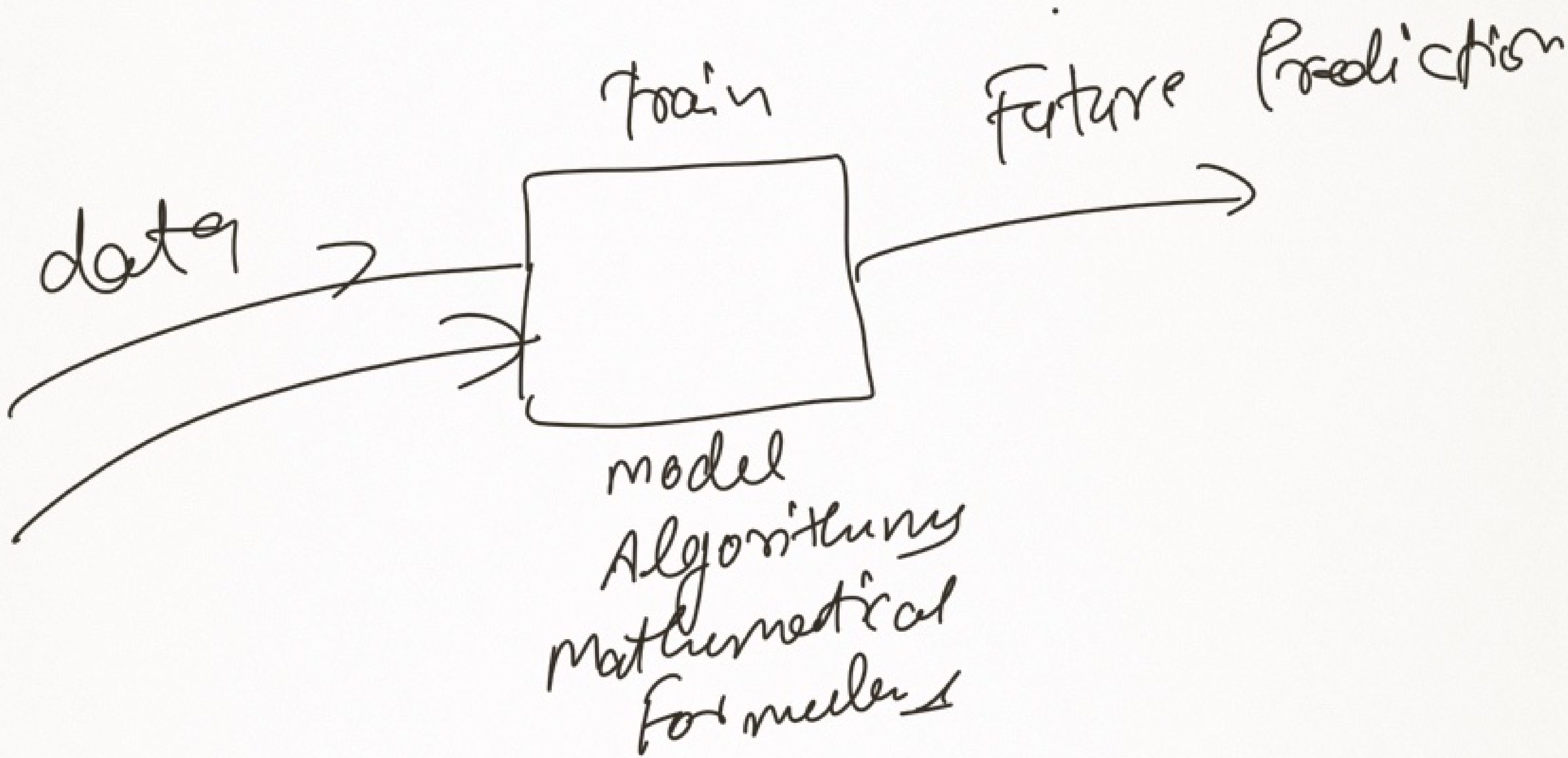
Machine Learning



Age	Salary
20	50K
21	60K
22	70K
23	80K
24	?

Age	Education	Tools
20	BTech	Python

mathematics

ML

Supervised | Unsupervised

Age	Experience	Salary
20	5	70K
22	2	40K
23	7	60K
24	3	?

Supervised ML

Y

output data
Predicted data
dependent data

Input Column
Independent Column
X

Age	Salary
20	50K
22	60K
23	70K
24	80K

Rich

Poor

Input collates

Unsupervised Lear

Reinforcement Learning

Supervised ML

Linear Reg

Sales

100.45 $

120.50 $

130.50 $

100 0 $

1000 $

Diabetes Prg. function

0.235	0
0.456	to
0.250	1
0.956	

Catagorical

Color

R → 0

→ 1

0

1

0

1

2

1

Contineor

Linear Regression

$Y = f(x)$ $\rightarrow Y = mx + c$

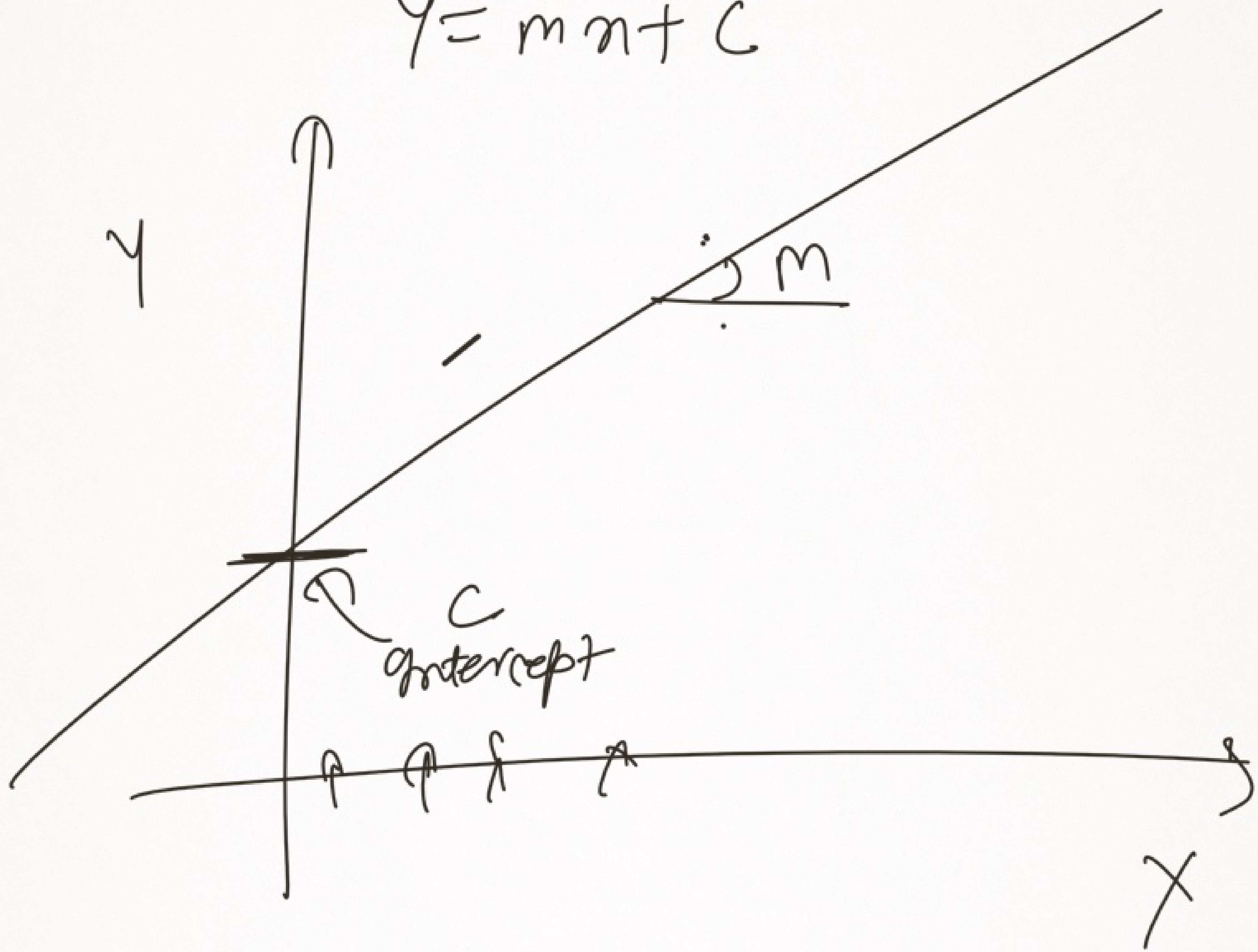
Indep

Age
$x = 25$

Let
$m = 1$
$c = 10$

$Y = 1 \times 25 + 10$
$Y = 25 + 10$
$Y = 35K$

$Y = mx + C$



X	Y act	Pred
1	2	2
2	4	2.5
3	6	3.5

0

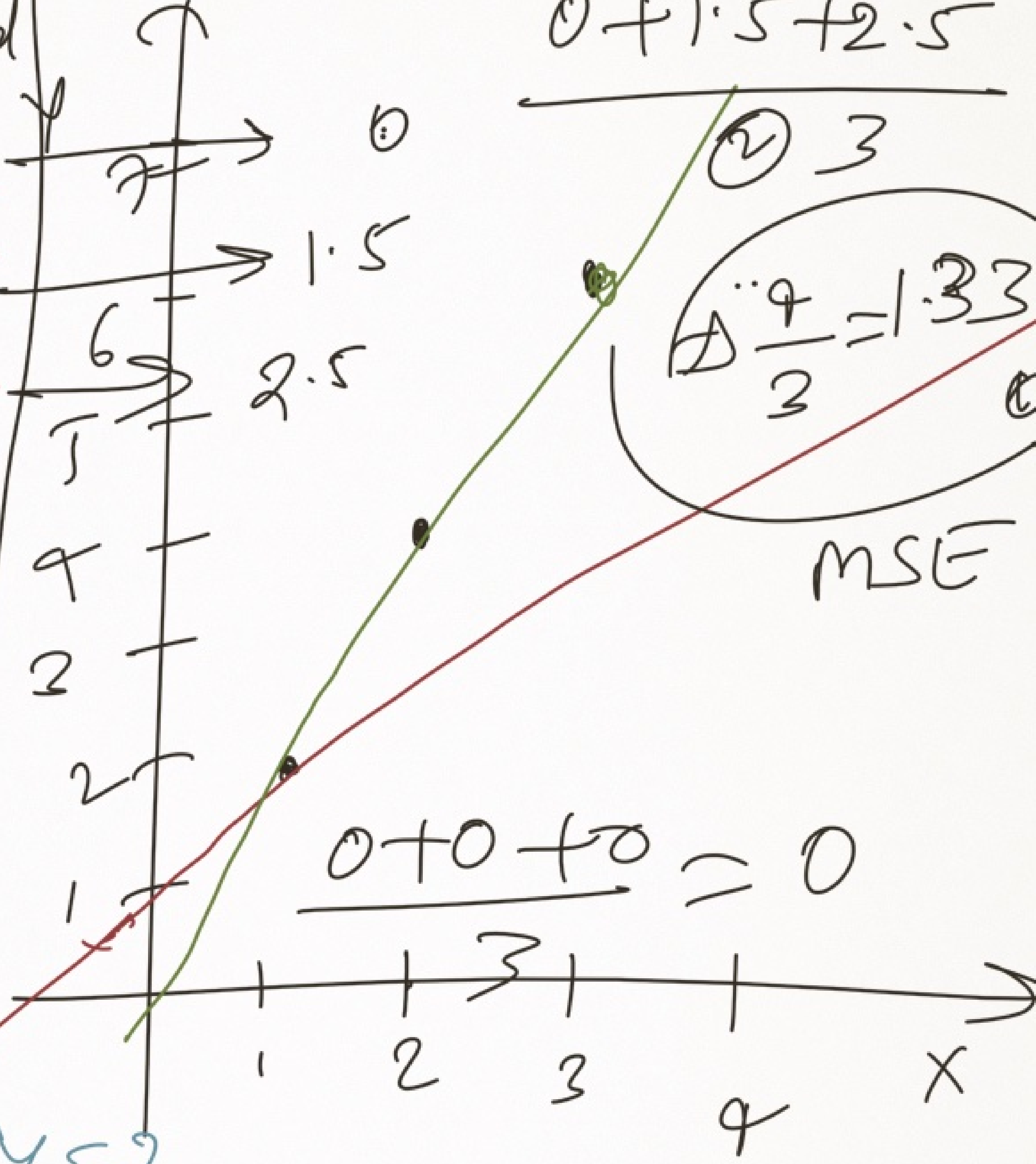
1.5

2.5

$$\frac{0+1.5+2.5}{3}$$

$$\frac{4}{3} = 1.33$$

MSE



pred	
2	0
4	0
6	0

$$\frac{0+0+0}{3} = 0$$

$n=4$ $Y=?$

pred 8

Y | $h_\theta(x)$ (Predicted)

① | -2

② | $+2$

③ | 0

$\Rightarrow \frac{-2+2+0}{3} = 0$

$Y = mx + c$

$h_\theta(x) = \theta_1 x + \theta_0$

$$\frac{\sum_{i=1}^{i=n} h_\theta(x)_i - Y_i}{n}$$

$\frac{2+2+0}{3} \rightarrow 1.33$

MAE

$(-2)^2$

$(+2)^2$

$(0)^2$

$\Rightarrow 1 \quad \frac{4+4+0}{3} = \frac{8}{3} \Rightarrow 2.6$

Squared over absolute

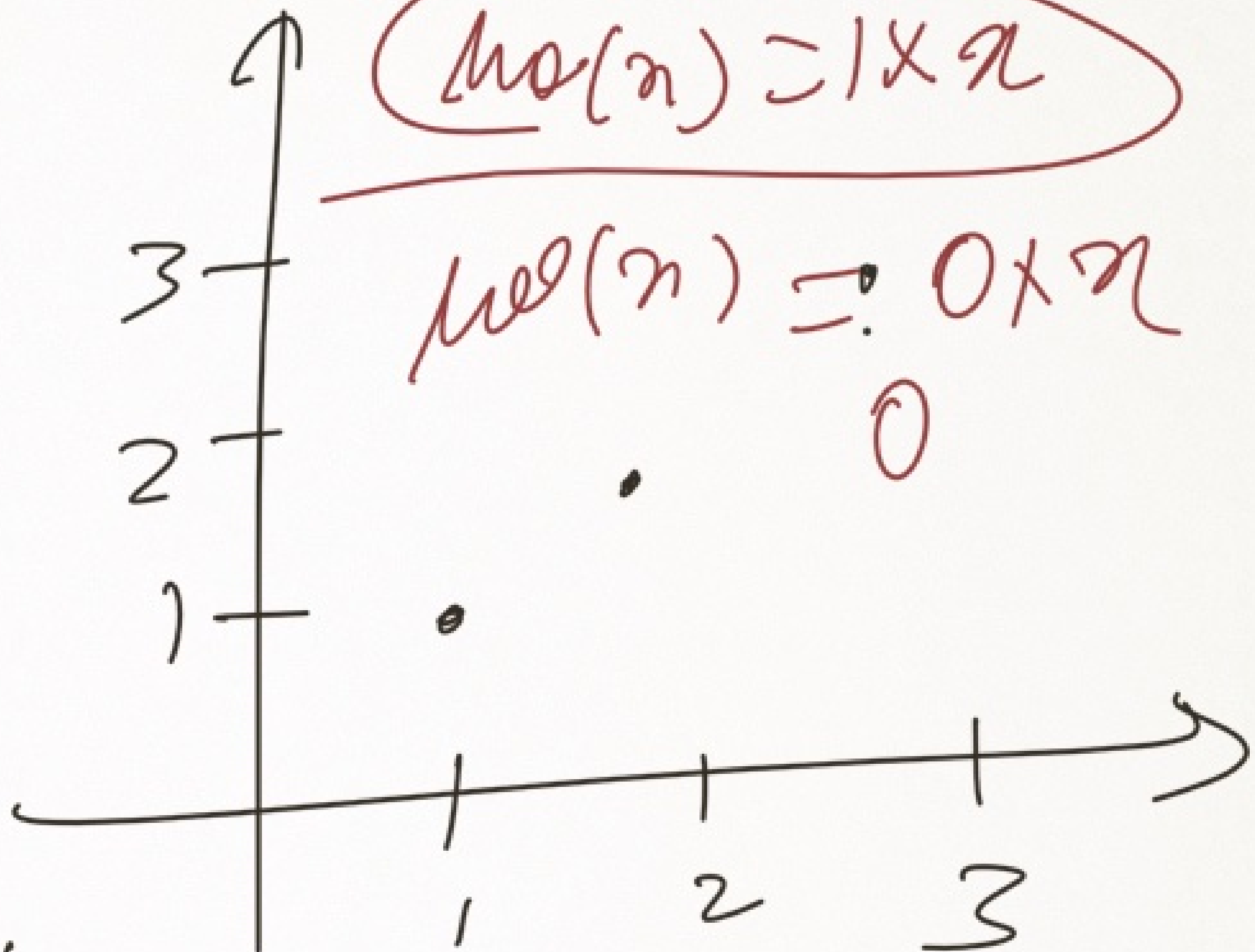
$$\frac{1}{2} \frac{\sum_{i=1}^{i=n} (h\theta(x)_i - y_i)^2}{n}$$

X	Y	$h_\theta(x)$
1	1	
2	2	
3	3	
4		4

$h_\theta(x) = 1 \times x$

$h_\theta(x) = 0 \times x$

0

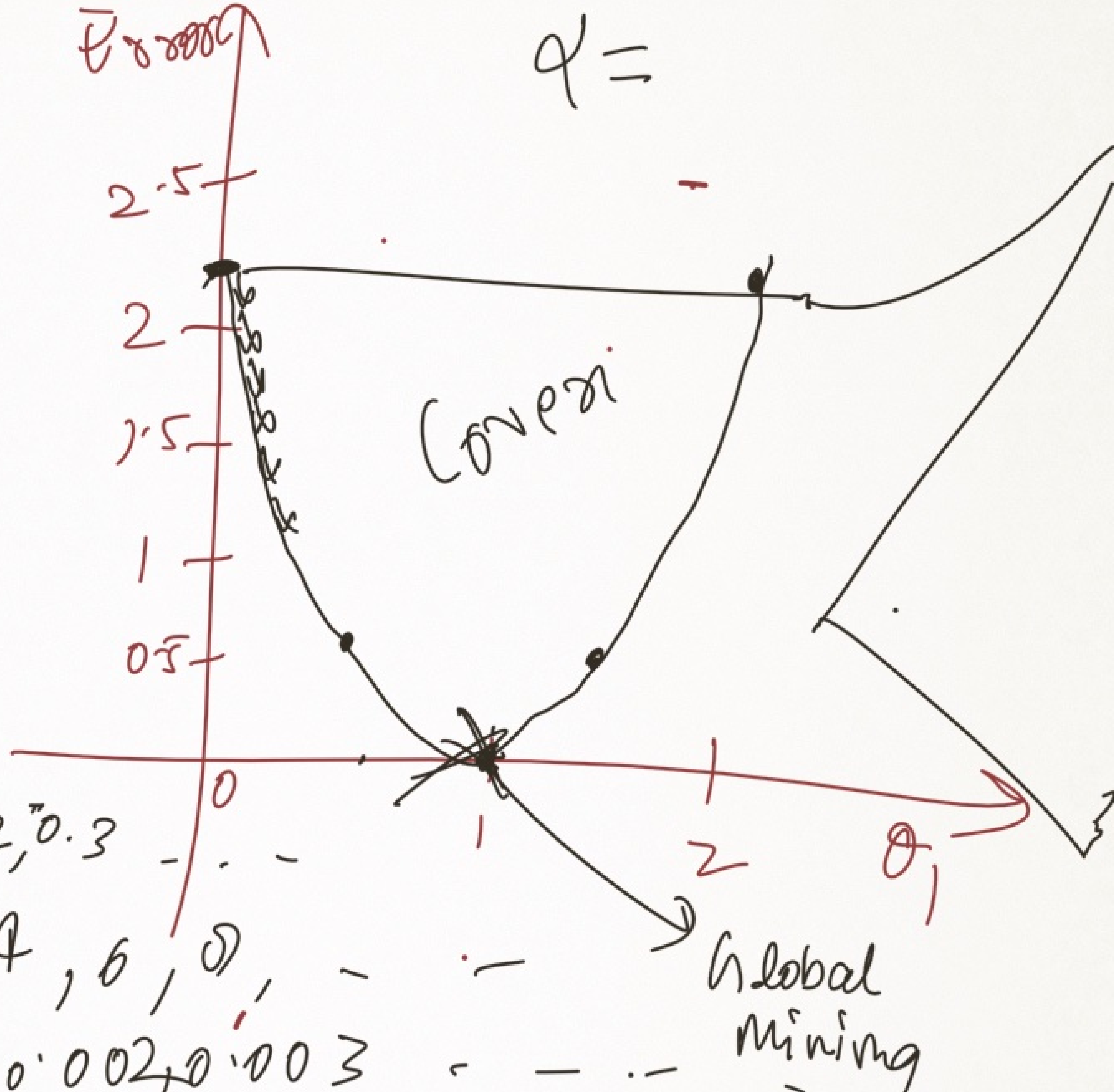


$$\frac{1}{2}\sum_{i=1}^{n}\frac{(h_\theta(x)_i - y_i)^2}{n}$$

$$\frac{(0.5-1)^2 + (1-2)^2 + (1.5-3)^2}{3}$$

θ_1	Error
0.5	0.58
0	2.33
1	0
1.5	0.58
2	2.33

θ_1	Error
0	2.33
0.5	0.58
1	0
1.5	0.58
2	2.33



① $\theta_1 = 0, 0.1, 0.2, 0.3$ - - -

② $\theta_1 = 0, 2, 4, 6, 8,$ - -

③ $\theta_1 = 0.001, 0.002, 0.003$ - - -

1, 2, 3, 4, 5

0.0 → 0.01, 0.02, 0.03, 0.09

10

10, 20, 30 . .

100

α

$$\theta_{1(new)} = \theta_{1}old - \underset{\uparrow \atop \text{LR}}{\alpha} \frac{\partial}{\partial \theta_1}\left[\text{Error}\right]$$

$$h_\theta(x) = \overset{(m)}{\theta_1} x + \overset{(c)}{\theta_0}$$

$$\theta_{0(new)} = \theta_0(old) - \alpha \frac{\partial}{\partial \theta_0}\left[\text{Error}\right]$$

$$\theta_{1(new)} = \theta_{1old} - \alpha \frac{\partial}{\partial \theta_1}\left[\frac{1}{2}\sum_{i=1}^{i=n}\frac{(h\theta(x)_i - y_i)^2}{n}\right]$$

$$\frac{1}{2n}\sum_{i=1}^{i=n}\frac{\partial}{\partial \theta_1}(\theta_1 x + \theta_0 - y)^2$$

$$\frac{1}{\cancel{2}n}\sum_{i=1}^{i=n}\cancel{2}(\theta_1 x + \theta_0 - y) \times x$$

$$\theta_{1(new)} = \theta_{1(old)} - \alpha\left[\sum_{i=n}^{i=n}\frac{(\theta_1 x + \theta_0 - y)\times x}{n}\right]$$

$$\theta_0(new) = \theta_0(old) - \alpha \frac{\partial}{\partial \theta_0}\left[\frac{1}{2n}\sum_{i=1}^{i=n}\left(h\theta(x)_i - y_i\right)^2\right]$$

$$\frac{1}{2n}\sum_{i=1}^{i=n}\frac{\partial}{\partial \theta_0}\left(\theta_1 x + \theta_0 - y\right)^2$$

θ_0

$$\cancel{2}\left(\theta_1 x + \theta_0 - y\right) \times 1$$

$$\theta_0(new) = \theta_0(old) - \alpha\left[\frac{1}{n}\sum_{i=1}^{i=n}\left(\theta_1 x + \theta_0 - y\right)\right]$$

$$\theta_1(\text{new}) = \theta_{1\,\text{old}} - \alpha\left[\frac{1}{n}\sum_{i=1}^{i=n}(\theta_1 x + \theta_0 - y)\times x\right]$$

$$\theta_0(\text{new}) = \theta_0(\text{old}) - \alpha\left[\frac{1}{n}\sum_{i=1}^{i=n}(\theta_1 x + \theta_0 - y)\times 1\right]$$